UNIVERSITAS GADJAH MADA

Metode Numeris

Minggu V: Solusi persamaan linear simultan dengan metode **Jacobi** dan **Gauss-Seidel**

Tujuan pembelajaran (LO)

- TEI124-LO1 Students are able apply root search problems from nonlinear equations by applying graphical methods, bracketing methods, and open methods to solve engineering problems.
- TEI124-LO2 Students are able to apply some methods such as Gauss, Gauss Jordan, Jacobi, Gauss Seidel, and LU Factorisation to solve engineering computational problems related to matrix theory such as matrix determinant calculation, matrix inverse, and simultaneous linear equations.
- TEI124-LO3 Students to apply the existing methods to solve engineering problems which is related to regression and interpolation.
- TEI124-LO4 Students are able to apply the Trapezoidal Method, Simpson Method, Romberg Integration Method, gauss quadrature Method to solve engineering problems related to numerical integral equations.
- TEI124-LO5 Students are able to apply methods such as Euler method, Heun method, Midpoint method, Runge-Kutta method, Shooting method to solve engineering problems related to ordinary differential equations equations.
- TEI124-LO6 Students are able to apply methods such as Finite Element method, and Finite Difference method to solve engineering problems related to partial differential equations.

Persamaan linear simultan

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n$$

Nilai x dapat dicari secara langsung atau melalui **iterasi**

$$AX = B$$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Metode Iterasi

- Prosedur dasar:
  - Selesaikan persamaan linear untuk x
  - Asumsikan nilai tebakan awal
  - Hitung nilai x
  - Lakukan langkah sebelumnya sampai eror/galat dapat ditoleransi
- Ada 2 metode yang akan dipelajari:
  - Metode iterasi Jacobi
  - Metode iterasi Gauss-Seidel

Metode Iterasi Jacobi

- Misal sistem 3 persamaan:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

- diselesaikan untuk x:

$$x_1 = \frac{(b_1 - a_{12}x_2 - a_{13}x_3)}{a_{11}}$$
$$x_2 = \frac{(b_2 - a_{21}x_1 - a_{23}x_3)}{a_{22}}$$
$$x_3 = \frac{(b_3 - a_{31}x_1 - a_{32}x_2)}{a_{33}}$$

- nilai x untuk iterasi ke-k (positif, selain 0):

$$x_1^k = \frac{\left(b_1 - a_{12}x_2^{k-1} - a_{13}x_3^{k-1}\right)}{a_{11}}$$
$$x_2^k = \frac{\left(b_2 - a_{21}x_1^{k-1} - a_{23}x_3^{k-1}\right)}{a_{22}}$$
$$x_3^k = \frac{\left(b_3 - a_{31}x_1^{k-1} - a_{32}x_2^{k-1}\right)}{a_{33}}$$

- x_1^0,x_2^0, x_3^0 adalah nilai awal
- iterasi berakhir ketika:

$$\left|\frac{x_i^k - x_i^{k-1}}{x_i^k}\right| 100\% < toleransi$$

Metode Iterasi Gauss-Seidel

- Misal sistem 3 persamaan:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$
$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

- diselesaikan untuk x:

$$x_1 = \frac{(b_1 - a_{12}x_2 - a_{13}x_3)}{a_{11}}$$
$$x_2 = \frac{(b_2 - a_{21}x_1 - a_{23}x_3)}{a_{22}}$$
$$x_3 = \frac{(b_3 - a_{31}x_1 - a_{32}x_2)}{a_{33}}$$

- nilai x untuk iterasi ke-k (positif, selain 0):

$$x_1^k = \frac{(b_1 - a_{12}x_2^{k-1} - a_{13}x_3^{k-1})}{a_{11}}$$
$$x_2^k = \frac{(b_2 - a_{21}x_1^k - a_{23}x_3^{k-1})}{a_{22}}$$
$$x_3^k = \frac{(b_3 - a_{31}x_1^k - a_{32}x_2^k)}{a_{33}}$$

- x_1^0, x_2^0, x_3^0 adalah nilai awal
- iterasi berakhir ketika:

$$\left|\frac{x_i^k - x_i^{k-1}}{x_i^k}\right| 100\% < toleransi$$

Contoh #1

- Selesaikan sistem persamaan berikut dengan metode iterasi:

$$3x + y - z = 5$$
$$4x + 7y - 3z = 20$$
$$2x - 2y + 5z = 10$$

- Langkah #1, selesaikan untuk x:

$$x = \frac{5 - y + z}{3}$$
$$y = \frac{20 - 4x + 3z}{7}$$
$$z = \frac{10 - 2x + 2y}{5}$$

x=1,506; y=3,1325; z=2,6505

$$x = \frac{5 - y + z}{3}$$

$$y = \frac{20 - 4x + 3z}{7}$$

$$z = \frac{10 - 2x + 2y}{5}$$

x=1,506; y=3,1325; z=2,6505

$$x = \frac{5 - y + z}{3}$$

$$y = \frac{20 - 4x + 3z}{7}$$

$$z = \frac{10 - 2x + 2y}{5}$$

Jebakan Metode Iterasi

- Solusi hanya akan konvergen jika matrix A memiliki komponen diagonal yang dominan:

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|$$

$$\cancel{\begin{bmatrix} & 5.81 & \\ 45 & & 1 \\ 1 & 16 & \end{bmatrix}} \qquad \begin{bmatrix} 124 & 34 & 56 \\ 23 & 53 & 5 \\ 86 & 34 & 129 \end{bmatrix}$$

- Sistem nyata yang dimodelkan biasanya memiliki komponen diagonal yang dominan

Kesimpulan

- Metode iterasi Jacobi dan Gauss-Seidel
- Perulangan dilakukan sampai eror/galat dapat ditoleransi
- Nilai awal
- Jebakan metode iterasi

PR

1. Suatu persamaan linear:

$$x + 8y - 2z = -9$$
$$-2x + 4y - 9z = -50$$
$$10x - 3y + 6z = 24{,}5$$

Hitung nilai x, y, dan z dengan menggunakan:

a. Metode eliminasi Gauss
b. Metode Gauss-Jordan
c. Metode iterasi Jacobi
d. Metode iterasi Gauss-Seidel

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Terima Kasih